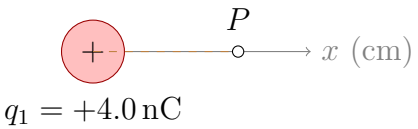
Electric Field Superposition Practice

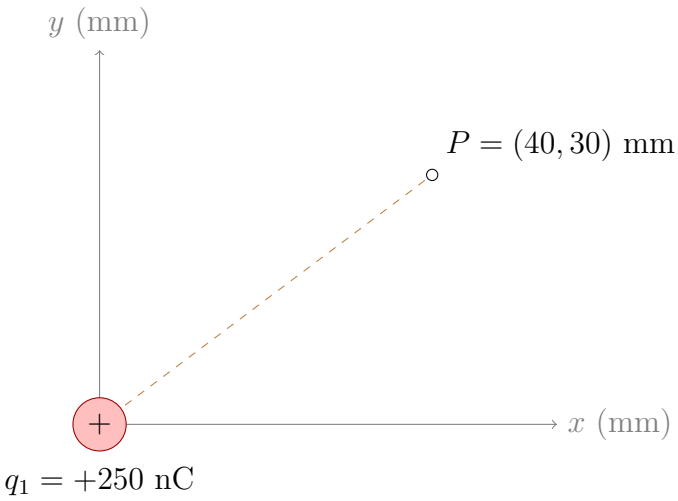
Part A — Separation Vectors to a Field Point

1. A charge $q_1 = +4.0\,\text{nC}$ sits at the origin. Find the vector $\vec{r}_1$ from q_1 to the point $P = (6.0\text{ cm}, 0)$, and its magnitude $|\vec{r}_1|$.



2. A charge $q_1 = -2.0\,\text{nC}$ sits at $(2.0\text{ cm}, 3.0\text{ cm})$. Find $\vec{r}_1$ from q_1 to the point $P = (6.0\text{ cm}, 6.0\text{ cm})$, and $|\vec{r}_1|$.

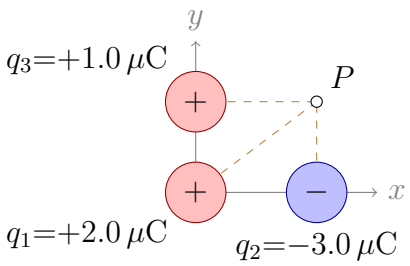
3. A charge $q_1 = +250$ nC sits at the origin. Find $\vec{r}_1$ from q_1 to the point $P = (40\text{ mm}, 30\text{ mm})$, and $|\vec{r}_1|$, in meters.



4. A charge $q_1 = -1.0\times10^{-6}$ C sits at $(5.0\times10^{-3}\text{ m}, 0)$. Find $\vec{r}_1$ from q_1 to the point $P = (0, 1.2\times10^{-2}\text{ m})$, and $|\vec{r}_1|$.

Part B — Separation Vectors, Multiple Charges to One Point

5. Three charges: $q_1 = +2.0\,\mu\text{C}$ at $(0, 0)$, $q_2 = -3.0\,\mu\text{C}$ at $(4.0\text{ cm}, 0)$, $q_3 = +1.0\,\mu\text{C}$ at $(0, 3.0\text{ cm})$. All three contribute to the field at $P = (4.0\text{ cm}, 3.0\text{ cm})$. Find $\vec{r}_1$, $\vec{r}_2$, and $\vec{r}_3$ (each charge to P), and their magnitudes.



6. Two charges: $q_1 = +150$ nC at $(0, 0)$, $q_2 = -80$ nC at $(30\text{ mm}, 0)$. Both contribute to the field at $P = (30\text{ mm}, 40\text{ mm})$. Find $\vec{r}_1$ and $\vec{r}_2$, and their magnitudes.

7. 3D. Two charges: $q_1 = +5.0\,\text{nC}$ at the origin $(0, 0, 0)$, $q_2 = -4.0\,\text{nC}$ at $(5.0\text{ cm}, 0, 0)$. Both contribute to the field at $P = (0, 4.0\text{ cm}, 3.0\text{ cm})$. Find $\vec{r}_1$ and $\vec{r}_2$ (each as an (x, y, z) triple), and their magnitudes.

8. 3D. Three charges: $q_1 = +90$ nC at $(1.0 \times 10^{-2}\ \text{m}, 0, 0)$, $q_2 = -60$ nC at $(0, 2.0 \times 10^{-2}\ \text{m}, 0)$, $q_3 = +45$ nC at $(0, 0, 2.0 \times 10^{-2}\ \text{m})$. All three contribute to the field at the origin $P = (0, 0, 0)$. Find $\vec{r}_1$, $\vec{r}_2$, and $\vec{r}_3$, and their magnitudes.

Part C — Electric Field Magnitude & Direction

Same four single-charge setups as Part A. Use $E = k|q|/r^2$, along with $\vec{r}_1$ from Part A, to get the field at P as a magnitude and a direction. (Careful with sign: the field of a negative charge points toward *it, not along $\vec{r}_1$.)*

9. Same as Problem 1: $q_1 = +4.0$ nC, P a distance $|\vec{r}_1| = 6.0$ cm away along $+x$. Find the magnitude and direction of $\vec{E}_1$ at P.

10. Same as Problem 2: $q_1 = -2.0$ nC, $\vec{r}_1 = (4.0, 3.0)$ cm. Find the magnitude and direction of $\vec{E}_1$ at P.

11. Same as Problem 3: $q_1 = +250$ nC, $\vec{r}_1 = (40, 30)$ mm. Find the magnitude and direction of $\vec{E}_1$ at P.

12. Same as Problem 4: $q_1 = -1.0 \times 10^{-6}$ C, $\vec{r}_1 = (-5.0, 12) \times 10^{-3}$ m. Find the magnitude and direction of $\vec{E}_1$ at P.

Part D — Individual Field Vectors — Components

Same setups as Part B. Find the individual field $\vec{E}_i$ due to each charge alone at P, in component form. Don't add them together yet.

13. Same as Problem 5: $q_1 = +2.0\,\mu\text{C}$ at $(0, 0)$, $q_2 = -3.0\,\mu\text{C}$ at $(4.0, 0)$ cm, $q_3 = +1.0\,\mu\text{C}$ at $(0, 3.0)$ cm; $P = (4.0, 3.0)$ cm. Find $\vec{E}_1$, $\vec{E}_2$, and $\vec{E}_3$ at P, in component form.

14. Same as Problem 6: $q_1 = +150$ nC at $(0, 0)$, $q_2 = -80$ nC at $(30, 0)$ mm; $P = (30, 40)$ mm. Find $\vec{E}_1$ and $\vec{E}_2$ at P, in component form.

15. 3D. Same as Problem 7: $q_1 = +5.0$ nC at $(0, 0, 0)$, $q_2 = -4.0$ nC at $(5.0\text{ cm}, 0, 0)$; $P = (0, 4.0, 3.0)$ cm. Find $\vec{E}_1$ and $\vec{E}_2$ at P, in component form.

16. 3D. Same as Problem 8: $q_1 = +90$ nC at $(1.0, 0, 0) \times 10^{-2}$ m, $q_2 = -60$ nC at $(0, 2.0, 0) \times 10^{-2}$ m, $q_3 = +45$ nC at $(0, 0, 2.0) \times 10^{-2}$ m; P at the origin. Find $\vec{E}_1$, $\vec{E}_2$, and $\vec{E}_3$ at P, in component form.

Part E — Net Field — Superposition

Add the individual fields from Part D as vectors to get the total field $\vec{E}_{net}$ at P. For the 2D setups, report components, magnitude, and direction. For the 3D setups, report components and magnitude only.

17. Using your answers from Problem 9, find $\vec{E}_{\text{net}}$ at P (components, magnitude, and direction).

18. Using your answers from Problem 10, find $\vec{E}_{\text{net}}$ at P (components, magnitude, and direction).

19. 3D. Using your answers from Problem 11, find $\vec{E}_{\text{net}}$ at P (components and magnitude only).

20. 3D. Using your answers from Problem 12, find $\vec{E}_{\text{net}}$ at P (components and magnitude only). This is the hardest one on the sheet — nice work getting here.

ANSWER KEY — FLIP THE PAGE OVER (OR STAND ON YOUR HEAD)

Part A — Separation Vectors to P

1. $\vec{r}_1 = (6.0, 0)$ cm; $|\vec{r}_1| = 6.0$ cm
2. $\vec{r}_1 = (4.0, 3.0)$ cm; $|\vec{r}_1| = 5.0$ cm
3. $\vec{r}_1 = (40, 30)$ mm; $|\vec{r}_1| = 50$ mm $= 0.050$ m
4. $\vec{r}_1 = (-5.0, 12) \times 10^{-3}$ m; $|\vec{r}_1| = 1.3 \times 10^{-2}$ m

Part B — Separation Vectors, Multiple Charges

5. $\vec{r}_1 = (4.0, 3.0)$cm, $|\cdot| = 5.0$cm $\cdot$ $\vec{r}_2 = (0, 3.0)$cm, $|\cdot| = 3.0$cm $\cdot$ $\vec{r}_3 = (4.0, 0)$cm, $|\cdot| = 4.0$cm
6. $\vec{r}_1 = (30, 40)$mm, $|\cdot| = 50$mm $\cdot$ $\vec{r}_2 = (0, 40)$mm, $|\cdot| = 40$mm
7. (3D) $\vec{r}_1 = (0, 4.0, 3.0)$cm, $|\cdot| = 5.0$cm $\cdot$ $\vec{r}_2 = (-5.0, 4.0, 3.0)$cm, $|\cdot| = 7.07$cm
8. (3D) $\vec{r}_1 = (-1.0, 0, 0)$cm, $|\cdot| = 1.0$cm $\cdot$ $\vec{r}_2 = (0, -2.0, 0)$cm, $|\cdot| = 2.0$cm $\cdot$ $\vec{r}_3 = (0, 0, -2.0)$cm, $|\cdot| = 2.0$cm

Part C — Field Magnitude & Direction

9. $E_1 = 1.0 \times 10^4$ N/C, along $+x$
10. $E_1 = 7.19 \times 10^3$ N/C, 36.9° below $-x$ (toward q_1)
11. $E_1 = 8.99 \times 10^5$ N/C, 36.9° above $+x$
12. $E_1 = 5.32 \times 10^7$ N/C, 67.4° below $+x$ (toward q_1)

Part D — Individual Fields at P

13. $\vec{E}_1 = (5.75, 4.32) \times 10^6$ N/C $\cdot$ $\vec{E}_2 = (0, -3.00 \times 10^7)$ N/C $\cdot$ $\vec{E}_3 = (5.62 \times 10^6, 0)$ N/C
14. $\vec{E}_1 = (3.24, 4.32) \times 10^5$ N/C $\cdot$ $\vec{E}_2 = (0, -4.50 \times 10^5)$ N/C
15. (3D) $\vec{E}_1 = (0, 1.44, 1.08) \times 10^4$ N/C $\cdot$ $\vec{E}_2 = (5.09, -4.07, -3.05) \times 10^3$ N/C
16. (3D) $\vec{E}_1 = (-8.09, 0, 0) \times 10^6$ N/C $\cdot$ $\vec{E}_2 = (0, 1.35, 0) \times 10^6$ N/C $\cdot$ $\vec{E}_3 = (0, 0, -1.01) \times 10^6$ N/C

Part E — Net Field at P

17. $\vec{E}_{\text{net}} = (1.14, -2.57) \times 10^7$ N/C; $|\vec{E}_{\text{net}}| = 2.81 \times 10^7$ N/C, 66.1° below $+x$
18. $\vec{E}_{\text{net}} = (3.24 \times 10^5, -1.80 \times 10^4)$ N/C; $|\vec{E}_{\text{net}}| = 3.24 \times 10^5$ N/C, 3.2° below $+x$
19. (3D) $\vec{E}_{\text{net}} = (5.09, 10.3, 7.74) \times 10^3$ N/C; $|\vec{E}_{\text{net}}| = 1.39 \times 10^4$ N/C
20. (3D) $\vec{E}_{\text{net}} = (-8.09, 1.35, -1.01) \times 10^6$ N/C; $|\vec{E}_{\text{net}}| = 8.26 \times 10^6$ N/C